

MONALISA DOKANIA

✉ msdokania@gmail.com | [in linkedin.com/in/monalisa-dokania](https://www.linkedin.com/in/monalisa-dokania) | github.com/msdokania

PROFESSIONAL SUMMARY

Founding Software Engineer with experience in full-stack development, scalable/reliable backend services, distributed systems, data-driven pipelines, and scalable cloud platform design. Curious & motivated learner, with a strong sense of ownership.

TECHNICAL SKILLS

Languages	Python, Java, Go, TypeScript, JavaScript, HTML, CSS
Frameworks / Distributed Syst	React, Node.js, tRPC, Flask, FastAPI, REST API, Spring Boot, Microservices
Cloud & DevOps	AWS, Kubernetes, Jenkins, Docker, Terraform, CI/CD, Kafka
Databases	PostgreSQL, NoSQL, Redis, MySQL, Elasticsearch, MongoDB, data modeling
Software Engineering	SDLC, Design Patterns, System Design, System Architecture, Agile
Machine Learning & AI	Feature engineering, ML model prototyping, AI-assisted workflows, LLM frameworks

TECHNICAL EXPERIENCE

Software Engineer BLUSVN Technologies	June 2024 – Present Scottsdale, AZ
• Led design and end-to-end development - from design to deployment - of a React based distributed water-delivery platform as a founding engineer, owning the full product development lifecycle from requirements to production deployment. • Developed and optimized 80+ backend APIs using TypeScript (tRPC) and PostgreSQL, including identity and authentication workflows. Improved data-fetching efficiency and reduced end-to-end latency by 40%. • Implemented high performance Golang microservices leveraging concurrent worker pools and horizontal scaling, to process 5K+ orders/sec at peak load, with reliable, asynchronous execution. • Built and maintained AWS production infrastructure with Terraform and optimized Kubernetes cluster resource management by transitioning from standard Cluster Autoscaler to Karpenter, resulting in faster node provisioning & less infrastructure overhead. • Enhanced the system's security posture and implemented AWS / EKS best practices, & security-first coding practices, reducing high-severity vulnerabilities CVEs (by 20%) & increasing cluster compliance (65% to 76%). • Built Kafka-based event-driven location workflows to support responsive UI experiences in order & delivery status tracking. • Wrote & executed unit tests, integration tests, and load tests (Python), improving system stability & reducing production issues. • Developed Python-based load testing scripts using Locust to stress-test the system under peak load, monitoring performance metrics through Prometheus and Grafana dashboards to identify bottlenecks and implement optimizations. • Collaborated closely with cross-functional developers, reviewed 200+ PRs, managed CI/CD pipelines, improved development & debugging processes through AI-assisted workflows and contributed to architecture/design discussions. • Mentored junior developers and interns, improving team productivity and engineering quality; acted as Scrum Master, and managed Agile sprints and JIRA tasks, fostering a culture of ownership and rapid problem-solving.	

Graduate Teaching Assistant (GTA/GSA) Arizona State University (SCAI)	May 2023 - May 2024 Tempe, AZ
• Supported Software Verification, Validation Testing coursework for large classes (900+ students total). • Guided students in unit testing, debugging, code reviews, software reliability and object-oriented design concepts.	

OTHER PROJECTS

Agent-Based AI System for Document & Log Analysis | *Python, LangChain, FastAPI, Redis RQ, Vector DB, OpenAI*

- Built a custom **multi-agent LLM analysis pipeline** (Planner → Retriever → Analyst → Critic → Reporter) to perform structured, evidence-backed document and log analysis, reducing hallucinations compared to single-prompt workflows.
- Implemented a RAG workflow for **semantic document retrieval** and raw log diagnostics using **FastAPI, Redis RQ, and Qdrant vector store**, enabling asynchronous analysis of large PDFs and logs with verification and confidence scoring.

EDUCATION

Master of Science, Computer Science Arizona State University Tempe, AZ	GPA: 4.0/4.0
Bachelor of Technology, Information Technology Motilal Nehru National Institute of Technology Allahabad, UP, India	CGPA: 7.59/10.0